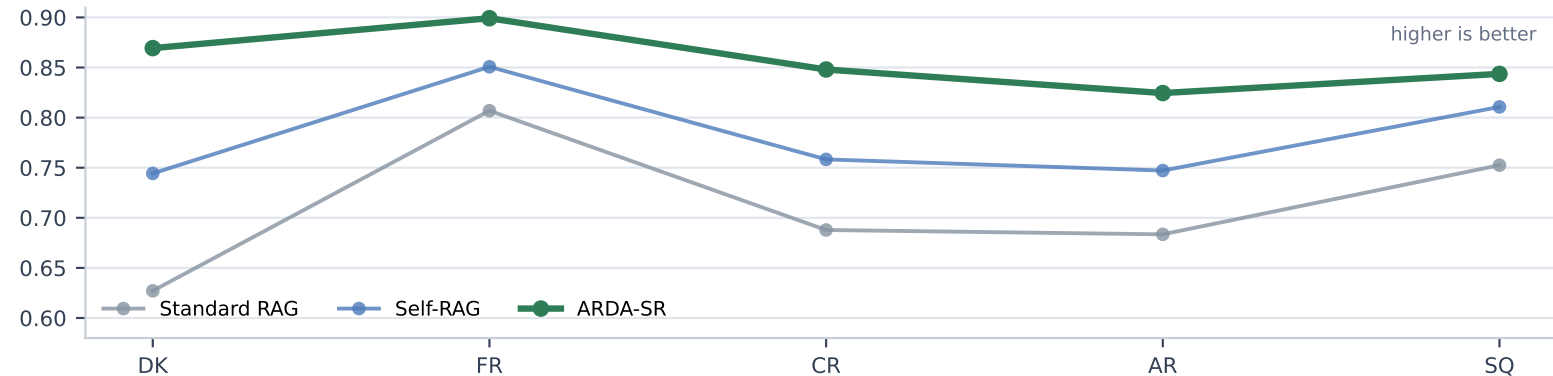


Per-Category Performance Comparison Between ARDA-SR and Representative Baselines

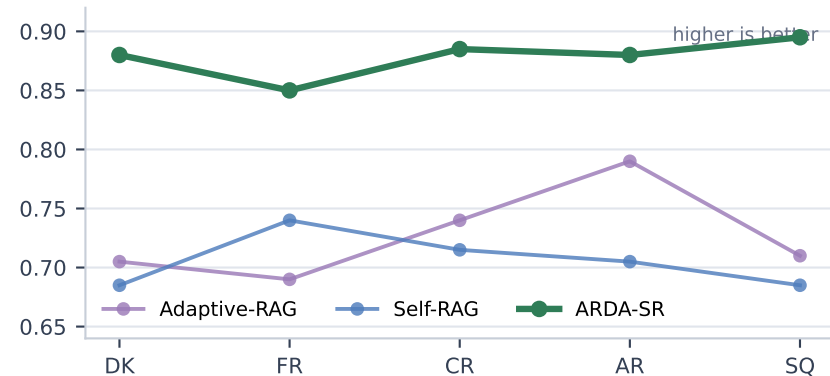
(a) Answer Quality: mean of Rel., Faith., Cov.



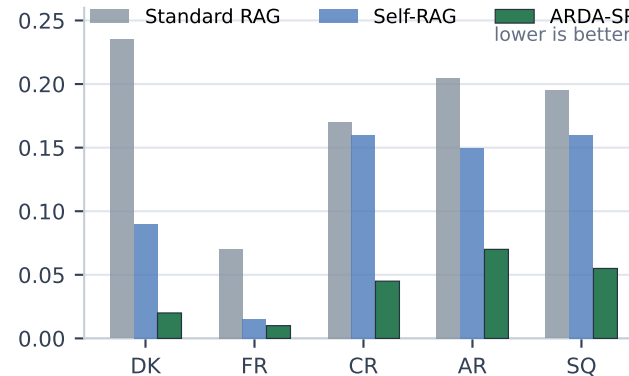
(b) Retrieval Quality: mean of Hit@5, CtxRel



(c) Reasoning & Decision: ToolAcc



(d1) Behavioral Risk: FRR



(d2) Efficiency: Latency



DK: Direct Knowledge FR: Factual Retrieval CR: Complex Reasoning AR: Ambiguous Routing SQ: Scenario Query

Pattern check: ARDA-SR leads answer quality across all QA categories and reduces FRR most strongly on CR, AR, and SQ; latency is the expected cost.